

Customer Churn Dataset — Analysis Report

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Customer Churn Dataset".

Dataset Overview

File	customer_churn_dataset (1).csv
Rows	2000
Columns	17
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	logistic_regression
Problem Type	classification
Target Column	churn

Metrics

Metric	Value
accuracy	0.73
precision	0.7211
recall	0.73
f1	0.7182
roc_auc	0.7762
confusion_matrix	{'labels': ['0', '1'], 'matrix': [[224, 35], [73, 68]]}

class_distribution	{'0': 259, '1': 141}
cv_scores	[0.7392043297649649, 0.71525, 0.7493493062198326, 0.7060372256031913, 0.6936931191395477]
cv_mean	0.7207

Feature Importance

Feature	Importance
contract_type_Month-to-Month	0.0019
contract_type_Two Year	0.0016
signup_date_2025-07-02	0.0016
signup_date_2026-01-07	0.0015
signup_date_2024-08-18	0.0014
customer_id_CUST11039	0.0014
customer_id_CUST10884	0.0014
signup_date_2023-12-23	0.0014
signup_date_2022-08-31	0.0013
customer_id_CUST11267	0.0013
signup_date_2025-01-01	0.0013
signup_date_2026-02-18	0.0013
signup_date_2025-06-27	0.0013
customer_id_CUST10323	0.0013
signup_date_2024-02-07	0.0013
days_since_last_purchase	0.0013
customer_id_CUST11912	0.0013
customer_id_CUST10715	0.0013
signup_date_2022-09-27	0.0013

signup_date_2024-09-05	0.0013
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AI Business Insight AI-generated insight

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.